

Shubham Inavolu

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EDUCATION

University of Massachusetts Amherst

Bachelor of Science - Computer Science

Achievements: Deans List, Presidential Service Award

Relevant Coursework: Object Oriented Programming, Data Structures, Design of Algorithms, Artificial Intelligence, Machine Learning, Web Programming

Amherst, MA

Graduation: July 2025

SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript

Web Development: React, Node.js, Express.js, Next.js, Tailwind CSS

Backend & Database: PostgreSQL, Firebase, AWS, Spring Boot, OAuth2

AI/ML: Pandas, NumPy, Scikit-learn

Testing & DevOps: JUnit, Jest, Azure, Jenkins, AWS Lambda, PM2, Docker, Kubernetes

EXPERIENCE

- Full Stack Developer Intern

June 2025 - August 2025

Staples

Framingham, MA

 - Developed a real-time Transportation Dashboard end-to-end using React, Next.js, Spring Boot, and Java to improve shipment visibility for the Supply Chain team, displaying 32 key insights and metrics for business users.
 - Collaborated with stakeholders to gather requirements, designed detailed wireframes in Figma, and structured the frontend with reusable components ready for backend integration.
 - Built RESTful APIs using Java to fetch PO numbers from an internal database, enrich shipment data via third-party APIs, and transform over 500 response objects per hour into optimized DTOs for efficient downstream use.
 - Engineered a time-based caching system to preload and store shipment data hourly, reducing frontend load times from 7 seconds to under 400 ms while maintaining scalability and responsiveness.
 - Integrated OAuth2 authentication to restrict access to authorized users, ensuring compliance with internal security protocols.
 - Containerized services with Docker, orchestrated deployments on Kubernetes (AKS), automated CI/CD pipelines with Jenkins, and deployed to Azure Cloud.
- Backend Developer

November 2024 - November 2024

Hack UMass Hackathon: Project Management Dashboard

Amherst, MA

 - Optimized backend services using Spring, implementing Redis caching to reduce database load and decrease API response times by 21% through improved service-layer performance and efficient data retrieval.
 - Developed and maintained RESTful APIs using Java to support scalable client-server interactions, ensuring reliable data exchange between the frontend and MySQL database.
 - Collaborated within an Agile team environment, contributing to sprint planning and peer code reviews while using GitHub for version control and continuous integration.
 - Integrated the JUnit testing framework into the development workflow, improving code quality and achieving 100% test coverage for core application logic and edge-case scenarios.

PROJECTS

- NavAIGate - Travel Planner (Live):

React, Node.js, Express.js, Firebase, Gemini AI, Vercel, Tailwind CSS

 - Built a microservices-based full-stack AI-powered trip planner using Gemini AI to generate personalized itineraries based on user preferences.
 - Integrated Google Places API and Firebase Authentication, scaling to 35+ users and 150+ AI-generated itineraries.
 - Reduced API response times by 30% using Axios caching and deployed on Vercel and Firebase Functions for scalability.
- MockMate - Interview Prep (Live):

Next.js, Node.js, Firebase, Gemini AI, Vapi, Tailwind CSS, Zod

 - Developed a full-stack web app for AI-powered mock interviews with real-time voice interaction using Vapi and intelligent question generation and evaluation via Gemini AI, resulting in over 40 interviews completed by users.
 - Integrated Firebase for secure user authentication with Google Auth and managed user data, transcripts, and feedback storage through Firestore, supporting over 20 registered users and 50 interview sessions.
 - Built personalized interview flows using voice-input preferences, validated with Zod, allowing users to generate interviews in under one minute, resulting in an estimated 50% boost in completion rates.
- Connect-4 AI:

Python, NumPy

 - Developed a Connect-4 AI using minimax with alpha-beta pruning, reducing evaluated states by 50%.
 - Optimized game state evaluations using vectorized NumPy operations, making AI computations 3x faster.
 - Implemented heuristics-based move prediction, achieving 70% accuracy in anticipating player strategies.